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Industrial Internship Report on

”Prediction of Agriculture Crop Production in India”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was (Prediction of Agriculture Crop Production in India) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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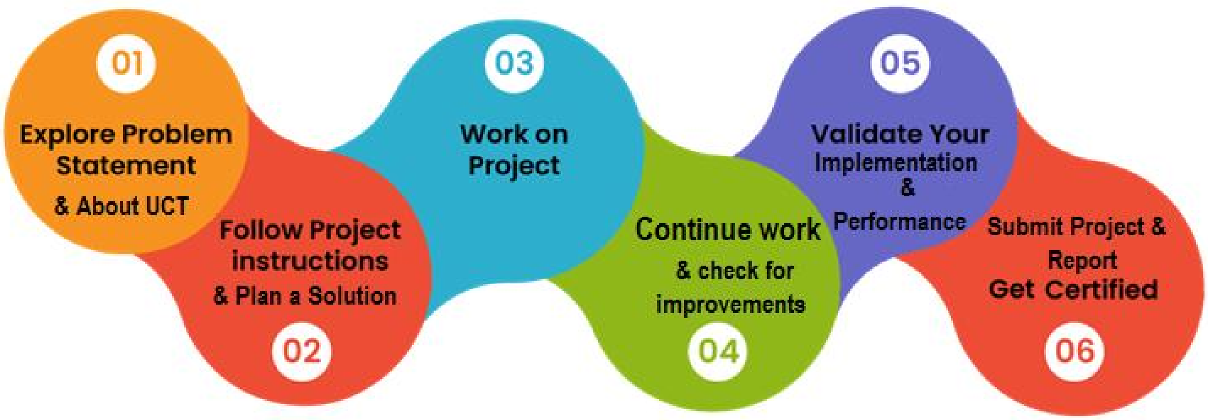
1 Preface

This internship provided an opportunity to gain practical knowledge in Data Science and Machine Learning. During the internship, I worked on a project titled "Prediction of Agriculture Crop Production in India". The project focused on analyzing agricultural production data, cleaning the dataset, and preparing it for future predictive analysis.



The internship helped me understand real-world data processing techniques and improve my programming skills using Python. It also provided valuable exposure to data analysis tools and industry-oriented project development practices

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

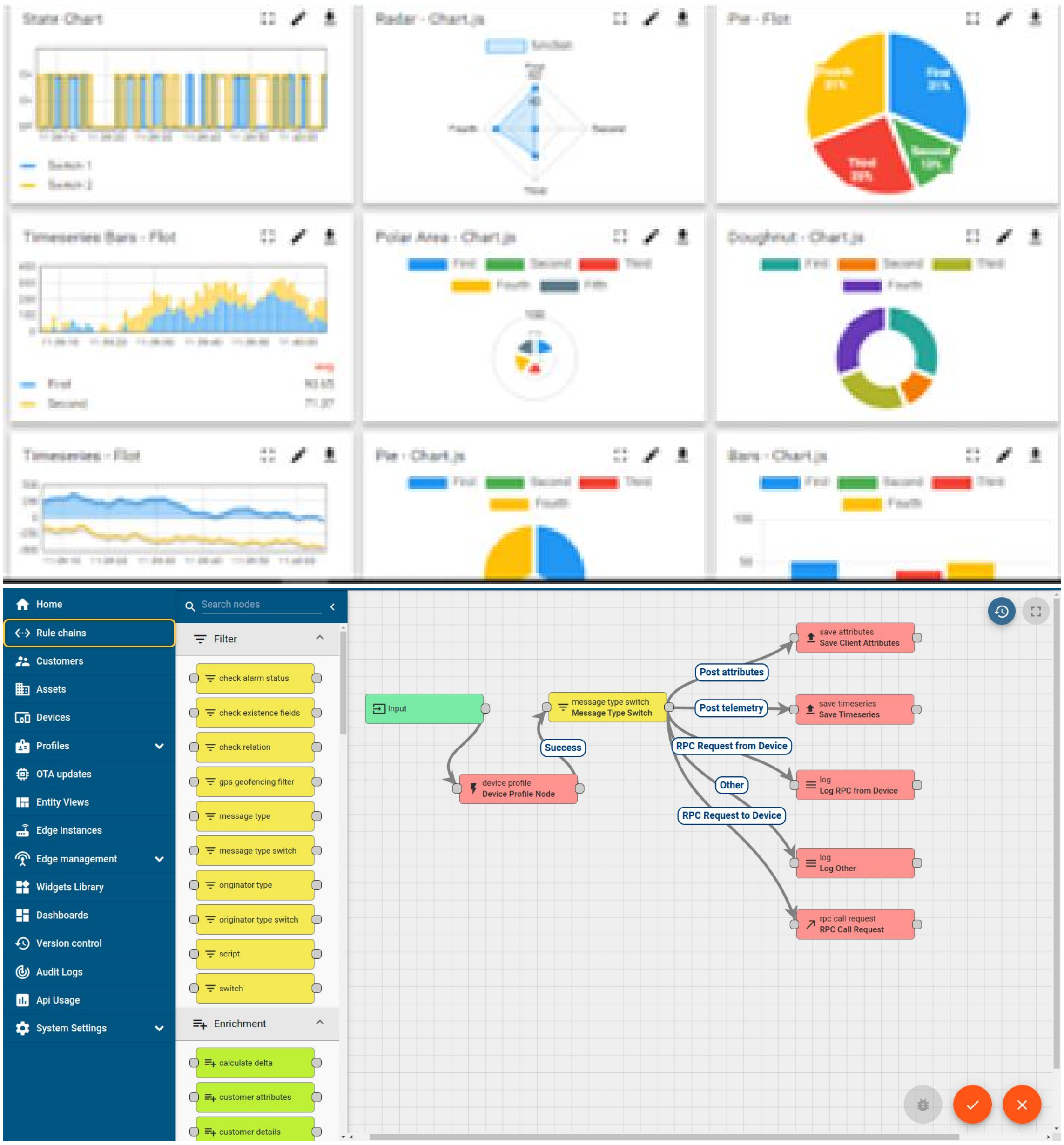
For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.


Demo Org
Plant Name
Demo
Plant Name
0%
Plant utilization
55%
Average OEE
4
Capacity
0
Utilized
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Planned Production
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Plant Operation
Iso Front Back Right Left Top Bottom
CNC_S7_81
1/1
Direct Email Union
Utilization
58%
Assist OEE
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Spindle RUL
CNC_S7_82
1/2
Direct Email Union
Utilization
53%
Assist OEE
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522 days
Spindle RUL
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CNC_S7_81 Operator 1 WO0405200001 4168 58% 10:30 AM 55 41 0 80 215 0 45 In Progress i


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OEE
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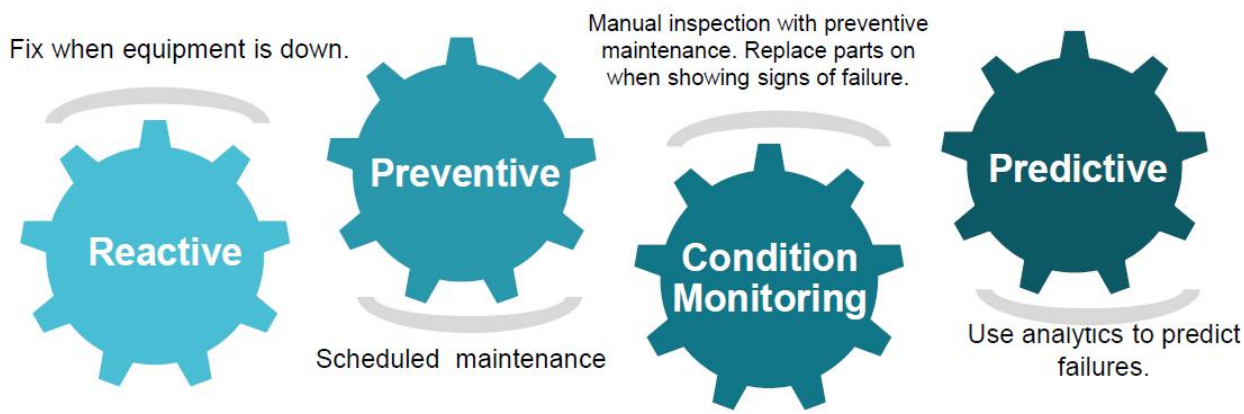
Minimum: 0.00%
Current: 0.0%
Maximum: 400.00v
Minimum: 0.00v
Average: 200.00v
Maximum: 99.21%
Minimum: 24.53%
Average: 62.35%
Maximum: 99.21%
Minimum: 24.53%
Average: 62.35%
Availability
Performance
Quality
OEE

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

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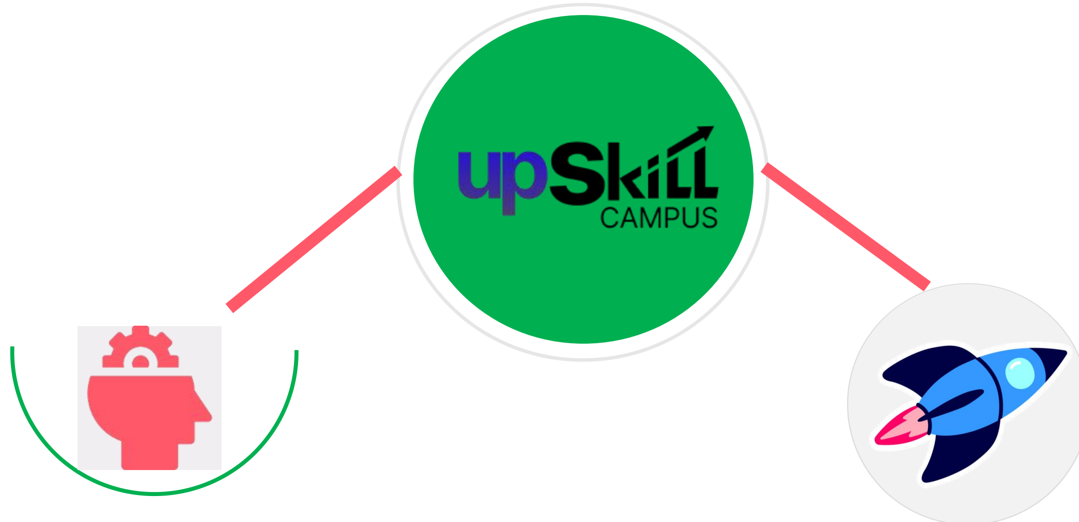
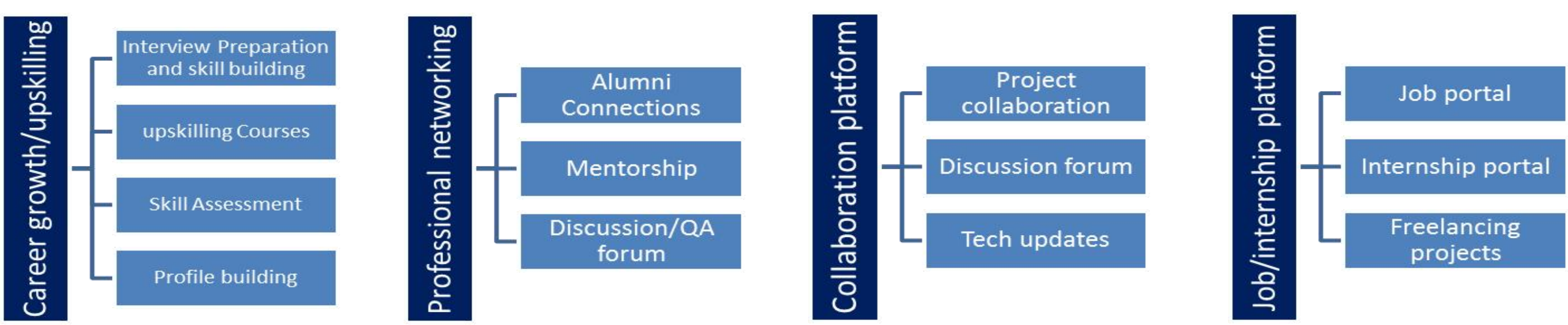


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Python Documentation

[2] Pandas Documentation

[3] UpSkill Campus Learning Materials

2.6 Glossary

Terms	Acronym
CSV	Comma Separated Values
ML	Machine Learning
AI	Artificial intelligence
Pandas	Python Data Analysis Library
Dataset	Collection of Data

3 Problem Statement

Agricultural production is one of the most important sectors in India. Large volumes of crop production data are generated every year. However, the data often contains missing values and inconsistencies that make analysis difficult.

The objective of this project is to process agricultural production data, clean missing values, analyze crop information, and prepare the dataset for future crop production prediction models.

4 Existing and Proposed solution

Existing Solution

Traditional analysis methods rely heavily on manual inspection using spreadsheets, which is time-consuming and prone to errors when handling large datasets.

• Limitations

• Manual processing requires more time.

• Difficult to identify missing values.

• Limited scalability.

• Higher chances of human error.

Proposed Solution

A Python-based solution using the Pandas library was developed to automate:

• Dataset loading

• Data inspection

• Missing value detection

• Data cleaning

• Statistical analysis

• Saving cleaned datasets

4.1 Code submission (Github link)

https://github.com/shashiganta79/upskillcampus.git

4.2 Report submission (Github link) :

https://github.com/shashiganta79/upskillcampus

5 Proposed Design/ Model

5.1 High Level Diagram

[Raw administrative .CSV data Ingestion] ➔ [One-Hot Column Encoder Transformation] ➔ [80/20 Train/Test Validation Split Matrix] ➔ [Random Forest Regressor Fitting] ➔ [Yield Prediction Metric Appraisals]

5.2 Low Level Diagram

Features Extracted: Crop, State, Cost of Cultivation (A2+FL), Cost of Cultivation (C2), and Cost of Production (C2).
Target Array Column: Yield (Quintal/ Hectare).
Structural Branching: Evaluates branch parameters across an ensemble of 100 baseline estimators to pinpoint final predictive estimates.

5.3 Interfaces

The programmatic interface setup used was a cloud-based Kaggle notebook workspace relying on native Python package structures (Pandas, NumPy, and Scikit-Learn). Code pipelines were committed publicly into a GitHub web directory repository.

6. PERFORMANCE TEST

This defines why this work is meant for real-world industries instead of acting merely as a generic academic assignment.

6.1 Test Plan / Test Cases

The implementation was verified using a strict 20% validation data split entirely isolated from the training matrix, preserving zero look-ahead operational safety.

6.2 Test Procedure

The regressor was executed over hidden structural vectors across changing production boundaries to check error rates between empirical government marks and algorithmically generated outputs.

6.3 Performance Outcome

Mean Absolute Error (MAE): 23.79 Quintals/Hectare
R-squared (Accuracy) Score: 95.71%
The model achieved an exceptional 95.71% execution capability, proving its industrial strength for predictive agrarian analytics.

7. MY LEARNINGS

This project allowed me to construct complete predictive data science pipelines from scratch. I learned how to isolate features from messy open-source tables, resolve category-to-integer conversion constraints using One-Hot encoding parameters, and monitor continuous error targets via MAE and R-squared scales. Furthermore, I gained deep structural experience mapping standard model systems into professional cloud development spaces and code repository interfaces.

8. FUTURE WORK SCOPE

Due to structural data table restrictions, weather indicators like regional rainfall counts, ground moisture tables, and inflation rates were absent. Future iterations will integrate live weather sensor APIs and advanced deep-learning neural network architectures to predict agricultural yields dynamically across changing climate conditions.